

Vol 9 Chapter 6 — Magnetism

Keeper (author pass — deep math/physics revision)

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Chapter 6 — Magnetism

Level 1 — one sentence

Magnetic phenomena — dia/para/ferro/antiferromagnetism — arise from electron spin and orbital angular momentum responding to applied fields, with exchange interactions (Heisenberg, Dzyaloshinskii-Moriya, anisotropy) producing the characteristic ordering temperatures (Curie, Néel) and order parameters; in BST these are substrate Pin(2) K-type-cluster alignments under exchange-coupling boundary conditions.

Level 2 — graduate-physicist precision

6.1 Types of magnetic response

- Diamagnetism: induced magnetic moment opposes applied field ($\chi < 0$). All materials have this; usually weak. Bismuth, water, organic compounds. Superconductors: $\chi = -1$ (perfect diamagnetism).
- Paramagnetism: aligned moments with applied field ($\chi > 0$, T -dependent). Curie law: $\chi = C/T$. Cu^{2+} , transition metal ions.
- Ferromagnetism: spontaneous parallel alignment below Curie temperature T_C . Fe ($T_C = 1043$ K), Co (1394 K), Ni (627 K), Gd (293 K). Hysteresis, domain structure.
- Antiferromagnetism: spontaneous antiparallel alignment below Néel temperature T_N . MnO ($T_N = 122$ K), Cr (308 K). Zero net magnetization.
- Ferrimagnetism: antiparallel sublattices with unequal moments. Net magnetization. Fe_3O_4 (magnetite), garnets, ferrites.

6.2 Heisenberg exchange

Quantum-mechanical exchange interaction between spins:

$$H_{\text{Heis}} = - \sum_{\langle i,j \rangle} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

$J > 0$: ferromagnetic (favors parallel). $J < 0$: antiferromagnetic. Originates from Coulomb exchange + Pauli exclusion.

Mean-field theory (Curie-Weiss): $\chi = C/(T - T_C)$ for ferromagnet above T_C . Below T_C : spontaneous magnetization $M(T) \propto (T_C - T)^{1/2}$ in mean-field; actual exponent ~ 0.32 in 3D Ising universality class (Vol 6 Ch 9).

6.3 Spin-wave excitations (magnons)

Ferromagnet's low-energy excitations: collective spin precessions called magnons (quantized spin waves). Dispersion $\omega \propto k^2$ for ferromagnets; $\omega \propto k$ for antiferromagnets (gapless Goldstone modes).

6.4 Magnetic anisotropy

Crystal-field effects + spin-orbit coupling produce direction-dependent magnetic response. Easy-axis: preferred magnetization direction. Anisotropy energy scale $\sim \mu\text{eV}$ - meV per atom for most materials.

Critical for permanent magnets (NdFeB, SmCo) — strong anisotropy holds magnetization against demagnetizing fields.

6.5 Frustration and spin liquids (Ch 11 preview)

Geometric frustration: triangular/kagome lattice antiferromagnets can't satisfy all $J < 0$ bonds simultaneously. Leads to spin liquid states with no magnetic order even at $T = 0$.

6.6 Worked example: iron magnetization

Iron (BCC, FCC at high T): $M_{\text{sat}}(0 \text{ K}) \sim 1750 \text{ kA/m}$, corresponds to $\sim 2.2 \mu_B$ per Fe atom. $T_C = 1043 \text{ K}$.

Above T_C : $\chi = C/(T - T_C)$ with $C \approx 2.4 \text{ emu}\cdot\text{K/mol}$. Paramagnetic susceptibility $\sim 10^{-4}$ at room T (well above magnetic Fe is paramagnetic).

6.7 Substrate Pin(2) reading

In BST: magnetic moments are substrate Pin(2)-charged K-type configurations. Exchange interactions couple substrate K-types via overlap of electronic K-type configurations. Ferromagnetic ordering = collective substrate K-type Pin(2)-alignment below T_C .

6.8 K-audit anchors

- Vol 5 Ch 3: substrate Pin(2) spin structure
- Vol 6 Ch 8: phase transitions framework (applies to magnetic ordering)

Level 3 — 5th-grader accessibility

Magnetism comes from electron spins and orbital motion. Five types: - Diamagnetism: weak repulsion from B field (all materials) - Paramagnetism: moments align with B but lose order at high T - Ferromagnetism: spontaneous alignment below Curie temperature (Fe, Ni, Co) - Antiferromagnetism: spontaneous anti-alignment (zero net magnetization) - Ferrimagnetism: anti-alignment but unequal — net magnetization (magnetite)

The Heisenberg exchange interaction $H = -J\vec{S}_i \cdot \vec{S}_j$ between neighboring spins drives ordering. Spin waves (magnons) are the collective excitations. Magnetic anisotropy picks easy axis. In BST: magnetism = substrate Pin(2) K-type alignment.

What comes next

Chapter 7 develops phonons.

Where to look this up

- Ashcroft and Mermin Ch 31-33
- Blundell, *Magnetism in Condensed Matter*
- BST: Vol 5 Ch 3